SQL Retail Sales Analysis

Data Exploration:-

-- How many sales we have?

Query:

```
SELECT
   COUNT(*) AS total_sale
FROM
   retail_sales;
```

Result: 1987

-- How many uniuque customers we have ?

Query:

```
SELECT
   COUNT(DISTINCT customer_id) AS total_sale
FROM
   retail_sales;
```

Result: 155

-- How many uniuque customers we have ?

Query:

```
SELECT DISTINCT category FROM retail_sales;
```

Result: Beauty, Clothing, Electronics

Data Analysis & Business Key Problems & Answers

My Analysis & Findings

Q.1 Write a SQL query to retrieve all columns for sales made on '2022-11-05
Q.2 Write a SQL query to retrieve all transactions where the category is 'Clothing' and the quantity sold is more than 10 in the month of Nov-2022
Q.3 Write a SQL query to calculate the total sales (total_sale) for each category.
Q.4 Write a SQL query to find the average age of customers who purchased items from the 'Beauty' category.
Q.5 Write a SQL query to find all transactions where the total_sale is greater than 1000.
Q.6 Write a SQL query to find the total number of transactions (transaction_id) made by each gender in each category.
Q.7 Write a SQL query to calculate the average sale for each month. Find out best selling month in each year
Q.8 Write a SQL query to find the top 5 customers based on the highest total sales
Q.9 Write a SQL query to find the number of unique customers who purchased items from each category.
Q.10 Write a SQL query to create each shift and number of orders (Example Morning <=12, Afternoon Between 12 & 17, Evening >17)

Q.1 Write a SQL query to retrieve all columns for sales made on '22-11-2022'

Query:

```
SELECT
   *
FROM
   retail_sales
WHERE
   sale_date = '2022-11-05';
```

Result:

Transaction id	Sale date	Sale time	Customer id	gender	age	category	qty	Price per unit	cogs	Total sale
180	2022-11-05	10:47:00	117	Male	41	Clothing	3	300	129	900
214	2022-11-05	16:31:00	53	Male	20	Beauty	2	30	8.1	60
240	2022-11-05	11:49:00	95	Female	23	Beauty	1	300	123	300
856	2022-11-05	17:43:00	102	Male	54	Electronics	4	30	9.3	120
943	2022-11-05	19:29:00	90	Female	57	Clothing	4	300	318	1200
1137	2022-11-05	22:34:00	104	Male	46	Beauty	2	500	145	1000
1256	2022-11-05	09:58:00	29	Male	23	Clothing	2	500	190	1000
1265	2022-11-05	14:35:00	86	Male	55	Clothing	3	300	111	900
1587	2022-11-05	20:06:00	140	Female	40	Beauty	4	300	105	1200
1819	2022-11-05	20:44:00	83	Female	35	Beauty	2	50	13.5	100
1896	2022-11-05	20:19:00	87	Female	30	Electronics	2	25	30.75	50

Q.2 Write a SQL query to retrieve all transactions where the category is 'Clothing' and the quantity sold is more than 4 in the month of Nov-2022

Query:

```
SELECT
   *
FROM
   retail_sales
WHERE
   category = 'Clothing'
      AND DATE_FORMAT(sale_date, '%Y-%m') = '2022-11'
      AND quantity >= 4;
```

Result:

Transaction id	Sale date	Sale time	Customer id	Gender	age	category	Qty	Price per unit	cogs	Total Sale
64	2022-11-15	06:34:00	7	Male	49	Clothing	4	25	8.5	100
146	2022-11-10	22:01:00	74	Male	38	Clothing	4	50	49	200
159	2022-11-10	21:30:00	42	Male	26	Clothing	4	50	23.5	200
284	2022-11-12	09:17:00	129	Male	43	Clothing	4	50	20.5	200
547	2022-11-14	07:36:00	3	Male	63	Clothing	4	500	250	2000

699	**2022-11-21**	**22:21:00**	**129**	**Female**	**37**	**Clothing**	**4**	**30**	**16.2**	**120**
735	**2022-11-26**	**21:38:00**	**153**	**Female**	**64**	**Clothing**	**4**	**500**	**515**	**2000**
943	**2022-11-05**	**19:29:00**	**90**	**Female**	**57**	**Clothing**	**4**	**300**	**318**	**1200**
965	**2022-11-27**	**21:45:00**	**84**	**Male**	**22**	**Clothing**	**4**	**50**	**13**	**200**
1259	**2022-11-03**	**17:31:00**	**105**	**Female**	**45**	**Clothing**	**4**	**50**	**21**	**200**
1296	**2022-11-26**	**20:42:00**	**45**	**Female**	**22**	**Clothing**	**4**	**300**	**342**	**1200**
1476	**2022-11-11**	**22:27:00**	**130**	**Female**	**27**	**Clothing**	**4**	**500**	**555**	**2000**
1484	**2022-11-23**	**09:29:00**	**22**	**Female**	**19**	**Clothing**	**4**	**300**	**147**	**1200**
1497	**2022-11-19**	**21:44:00**	**109**	**Male**	**41**	**Clothing**	**4**	**30**	**32.4**	**120**
1615	**2022-11-17**	**13:43:00**	**82**	**Female**	**61**	**Clothing**	**4**	**25**	**13.5**	**100**
1696	**2022-11-21**	**17:59:00**	**24**	**Female**	**50**	**Clothing**	**4**	**50**	**55**	**200**
1885	**2022-11-09**	**07:32:00**	**148**	**Female**	**52**	**Clothing**	**4**	**30**	**10.8**	**120**

Q.3 Write a SQL query to calculate the total sales (total_sale) for each category.

Query:

```
SELECT
   category,
   SUM(total_sale) AS net_sale,
   COUNT(*) AS total_orders
FROM
   retail_sales
GROUP BY 1;
```

Result:

Category	**net_sale**	**total_orders**
Beauty	**286790**	**611**
Clothing	**309995**	**698**
Electronics	**311445**	**678**

Q.4 Write a SQL query to find the average age of customers who purchased items from the 'Beauty' category.

Query:

```
SELECT
   ROUND(AVG(age), 2) AS avg_age
FROM
   retail_sales
WHERE
   category = 'Beauty';
```

Result: 40.42

Q.5 Write a SQL query to find all transactions where the total_sale is greater than 1000.

Query:

```
SELECT
  *
FROM
  retail_sales
WHERE
  total_sale > 1000;
```

Result:

Transaction id	Sale date	Sale time	Customer id	gender	age	category	Qty	Price per unit	cogs	Total sale
13	2023-02-08	17:43:00	106	Male	22	Electronics	3	500	245	1500
15	2022-07-01	11:50:00	75	Female	42	Electronics	4	500	210	2000
16	2022-06-25	10:33:00	82	Male	19	Clothing	3	500	180	1500
31	2023-12-31	17:47:00	3	Male	44	Electronics	4	300	129	1200
-	-	-	-	-	-	-	-	-	-	-
-	-	-	-	-	-	-	-	-	-	-
1956	2023-06-01	20:40:00	62	Male	30	Clothing	3	500	170	1500
1970	2023-05-22	17:44:00	5	Male	59	Electronics	4	500	230	2000

Q.6 Write a SQL query to find the total number of transactions (transaction_id) made by each gender in each category.

Query:

```
SELECT
  category,
  gender,
  COUNT(*) as total_trans
FROM retail_sales
GROUP
  BY
  category,
  gender
ORDER BY 1;
```

Result:

Category	Gender	Total trans
Beauty	Female	330
Beauty	Male	281
Clothing	Female	347
Clothing	Male	351
Electronics	Female	335
Electronics	Male	343

Q.7 Write a SQL query to calculate the average sale for each month. Find out best selling month in each year

Query:

```
SELECT
   year,
   month,
   round(avg_sale, 2) AS avg_sale
FROM (
   SELECT
      YEAR(sale_date) AS year,
      MONTH(sale_date) AS month,
      AVG(total_sale) AS avg_sale,
      RANK() OVER (PARTITION BY YEAR(sale_date) ORDER BY AVG(total_sale) DESC) AS r
   FROM retail_sales
   GROUP BY YEAR(sale_date), MONTH(sale_date)
) AS t1
WHERE r = 1;
```

Result:

year	month	avg_sale
2022	7	541.34
2023	2	535.53

Q.8 Write a SQL query to find the top 5 customers based on the highest total sales

Query:

```
SELECT
   customer_id, SUM(total_sale) AS total_sales
FROM
   retail_sales
GROUP BY 1
ORDER BY 2 DESC
LIMIT 5;
```

Result:

customer_id	total_sales
3	38440
1	30750
5	30405
2	25295
4	23580

Q.9 Write a SQL query to find the number of unique customers who purchased items from each category.

Query:

```
SELECT
   category, COUNT(DISTINCT customer_id) AS cnt_unique_cs
FROM
   retail_sales
GROUP BY category;
```

Result:

category	cnt_unique_cs
Beauty	141
Clothing	149
Electronics	144

Q.10 Write a SQL query to create each shift and number of orders (Example Morning <12, Afternoon Between 12 & 17, Evening >17)

Query:

```
WITH hourly_sale
AS
(
SELECT *,
   CASE
      WHEN EXTRACT(HOUR FROM sale_time) < 12 THEN 'Morning'
      WHEN EXTRACT(HOUR FROM sale_time) BETWEEN 12 AND 17 THEN 'Afternoon'
      ELSE 'Evening'
   END as shift
FROM retail_sales
)
SELECT
   shift,
   COUNT(*) as total_orders
FROM hourly_sale
GROUP BY shift;
```

Result:

shift	total_orders
Evening	1062
Morning	548
Afternoon	377

End of project